

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB

III Semester									
Course Code	Category	Hours / Week			Credits	Maximum Marks			
A6EE61	ESC	L	T	P	C	CIA	SEE	Total	
		0	0	2	1	40	60	100	
LIST OF EXPERIMENTS									
PART A : ELECTRICAL									
1	Identification, Specifications, Testing of R, L, C Components (Color Codes), Bread Boards, Diodes, BJTs. Verification of Ohms law								
2	Verification of KCL and KVL.								
3	Magnetization characteristics of DC shunt generator (Determination of critical field resistance and critical speed)								
4	Swinburne's test on dc shunt motor (Predetermination of efficiencies)								
5	OC and SC tests on Single-phase transformer.								
6	Brake test on 3-phase Induction motor.								
	PART B : ELECTRONICS								
7	Study and operation of Function Generator Regulated Power Supply (RPS) CRO.								
8	PN Junction Diode Characteristics (Forward bias, Reverse bias)								
9	Zener Diode Characteristics								
10	Rectifier without Filters (Full wave & half wave).								
11	Rectifier with Filters (Full wave & half wave).								
12	Transistor CE Characteristics (Input and Output)								
Text Books:									
1. Basic Electrical Engineering by <i>M.S.Naidu and S.Kamakshaiah</i> TMH 2. Electronic Devices and circuits by <i>J.Millman, C.C.Halkias and Satyabrata Jit</i> 2ed.,									
Reference Books:									
1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, "Basic Electrical, Electronics and Computer Engineering", Tata McGraw Hill, Second Edition, (2006). 2. Nagsarkar T K and Sukhija M S, "Basics of Electrical Engineering", Oxford press (2005). 3. Mehta V K, "Principles of Electronics", S.Chand & Company Ltd, (1994). 4. Mahmood Nahvi and Joseph A. Edminister, "Electric Circuits", Schaum' Outline Series, McGraw Hill, (2002).									

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

Student should able to

- 1 To analyze basic concepts to electric circuits
- 2 To apply electrical fundamentals to real time applications.
- 3 To apply electronics components to electronics circuits.
- 4 To create circuits containing basic electrical elements.
- 5 To apply electrical and electronics engineering concepts to real time application